Exploring House Prices in Ames, Iowa

Joudi Zakri | ML Foundations Capstone Report

1. Introduction

For this project, I used the Ames Housing dataset, which contains detailed information about residential house sales in Ames, Iowa. The dataset includes a wide range of features describing each property, such as size, quality, location, and sale price.

The goal of this project was to answer three main questions:
(1) What factors make a house more expensive?
(2) Which features are most strongly related to sale price?
(3) Are there any hidden patterns in the data?

2. Cleaning Summary

Before starting the analysis, the dataset required several cleaning steps:

• **Missing values:** Some columns such as *Alley, Pool QC, Fence,* and *Misc Feature* had a large number of missing values, so they were removed because they would not provide reliable information. For numeric columns like *Lot Frontage*, missing values were filled using the median, as it is less affected by extreme values.

• **Data types:** The *Lot Frontage* column was converted to numeric format to ensure correct calculations. Additionally, some columns were reviewed to confirm they had appropriate data types.

• **Outliers:** A boxplot of *SalePrice* showed extreme high values. Instead of removing them, I capped the values at the 99th percentile to reduce their impact while keeping the data realistic.

• **Duplicates:** Duplicate rows were checked and removed if found to avoid bias in the analysis.

All cleaning steps were organized inside a clean_data() function. I also added checks to ensure that sale prices are positive and that the dataset contains no critical missing values.

3. Feature Engineering Summary

To improve the analysis, I created several new features:

• **price_per_sqft = SalePrice / Gr Liv Area**
This feature normalizes the price by the size of the house, making comparisons more meaningful.

• **quality_area = Overall Qual × Gr Liv Area**
This interaction feature captures the combined effect of size and quality on price.

• **age = Yr Sold − Year Built**
This represents how old the house is at the time of sale.

• **age_group**
Houses were grouped into categories (New, Recent, Old) to simplify analysis.

• **Log transformation of SalePrice**
The original distribution was skewed, so applying a log transformation made it more balanced and suitable for analysis.

• **Encoding categorical variables**
Columns such as *Neighborhood* and *House Style* were converted into numerical format using one-hot encoding.

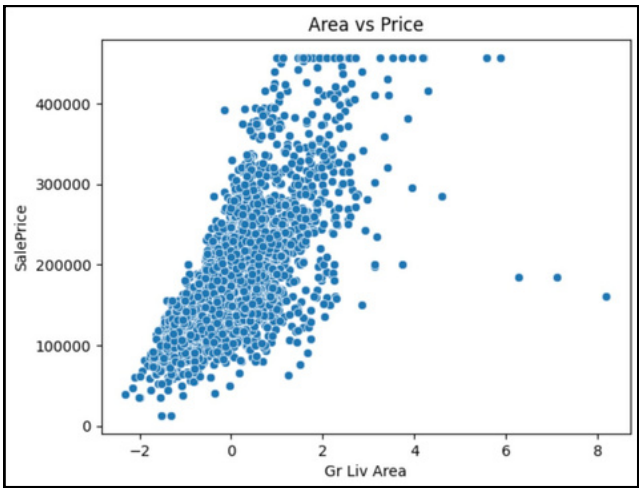
• **Scaling**
Some numerical features were standardized to improve consistency.

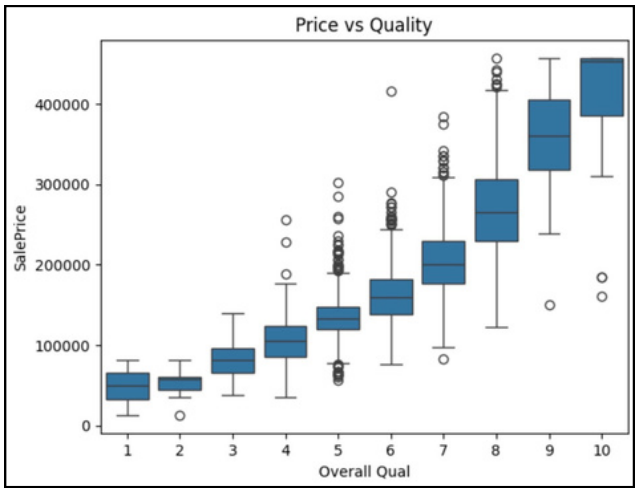
4. Key Findings

Finding 1 — Overall Quality has the strongest impact on price.
The analysis showed that houses with higher quality ratings have significantly higher prices. The boxplot clearly shows that prices increase as quality increases.

Finding 2 — Living area strongly affects price.
There is a clear positive relationship between house size (*Gr Liv Area*) and sale price. Larger houses tend to be more expensive, although the increase becomes less sharp for very large houses.

Finding 3 — Newer houses are generally more expensive.
The group analysis based on *age_group* showed that newer houses have higher average prices compared to older ones.





5. What I Would Do Next

If I had more time, I would:

(1) Build a machine learning model to predict house prices using the engineered features.
(2) Explore additional interaction features between location and house quality.
(3) Analyze trends over time to see how house prices changed across years.